DEEPEX

Deep Research Agent

Powered by Gemini API & Firecrawl

Project Technical Report

March 2026

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1. Project Overview

DEEPEX is an intelligent deep research agent that leverages Google's Gemini API and Firecrawl's advanced web-scraping capabilities to perform comprehensive, multi-step research on any topic. The agent autonomously searches the web, extracts and synthesizes relevant information, and generates an enhanced research report enriched with deeper insights, context, and analysis — all through a clean, interactive Streamlit interface.

The project bridges the gap between raw web data and structured research by combining two powerful AI-driven tools: Firecrawl for deep web research and content extraction, and Google's Gemini SDK for intelligent report generation and elaboration.

2. Features

- **Deep Web Research**: Automatically searches the web, extracts content, and synthesizes findings across multiple sources.
- **Enhanced Analysis**: Uses Google's Gemini SDK to elaborate on research findings with additional context, examples, and insights.
- **Interactive UI**: Clean Streamlit interface for easy interaction — no command-line expertise required.
- **Downloadable Reports**: Export research findings as markdown files for offline use or sharing.
- **Multi-step Pipeline**: A structured 3-phase pipeline ensures thorough coverage: research, synthesis, and enhancement.
- **Cited References**: The generated reports include citations and references sourced from real web pages.

3. System Architecture & How It Works

DEEPEX follows a well-defined 5-phase pipeline to deliver high-quality research reports:

1. Phase 1 – Input

The user provides a research topic and API credentials (Gemini API key and Firecrawl API key) via the Streamlit UI.

2. Phase 2 – Research

The tool uses Firecrawl's deep research endpoint to search the web and extract relevant information from multiple web sources. This step performs multiple iterations of searches and content extraction.

3. Phase 3 – Initial Synthesis

An initial research report is generated based on the Firecrawl findings using the Gemini API. This report is structured with sections such as introduction, literature overview, and analysis.

4. Phase 4 – Enhancement

A second Gemini agent elaborates on the initial report, adding depth, case studies, examples, practical implications, and comprehensive references.

5. Phase 5 – Output

The enhanced report is presented to the user in the Streamlit UI and made available for download as a markdown file.

4. Technical Requirements

Requirement	Details
Python Version	Python 3.8 or higher
Gemini API Key	Required – Google Generative AI access
Firecrawl API Key	Required – Firecrawl deep research endpoint
Python Packages	See requirements.txt (streamlit, google-generativeai, firecrawl-py, etc.)
Interface	Streamlit web application (runs locally)

5. Usage Guide

Step 1: Run the Streamlit app: `streamlit run deep_research_gemini.py`

Step 2: API Keys are automatically loaded from your `.env` file. Ensure the file is properly configured with your Gemini and Firecrawl API keys.

Step 3: Enter your research topic in the main input field.

Step 4: Click **Start Research** and wait for the 3-step process to complete.

Step 5: View and download your enhanced research report.

6. Example Research Topics

- Latest developments in quantum computing
- Impact of climate change on marine ecosystems
- Advancements in renewable energy storage
- Ethical considerations in artificial intelligence
- Emerging trends in remote work technologies
- Efficient Crypto primitive for Low resources devices security using LWC

7. Technical Details

The application employs two specialized AI agents:

Agent 1 – Research Agent

Utilizes Firecrawl's deep research endpoint to gather comprehensive information from multiple web sources. The Firecrawl deep research tool performs multiple iterations of web searches, content extraction, and analysis to provide thorough coverage of the topic. It retrieves real URLs and citations to ensure the research is grounded in verifiable sources.

Agent 2 – Elaboration Agent

Powered by Google's Gemini API, this agent enhances the initial research report by adding detailed explanations, examples, case studies, and practical implications. It produces a publication-quality document with structured sections, in-text citations, and a comprehensive reference list.

8. Sample Run – Results

The following screenshots demonstrate DEEPEX in action, using the research topic: *"Efficient Crypto primitive for Low resources devices security using LWC"*. The three-phase pipeline is shown step by step, followed by excerpts from the generated report.

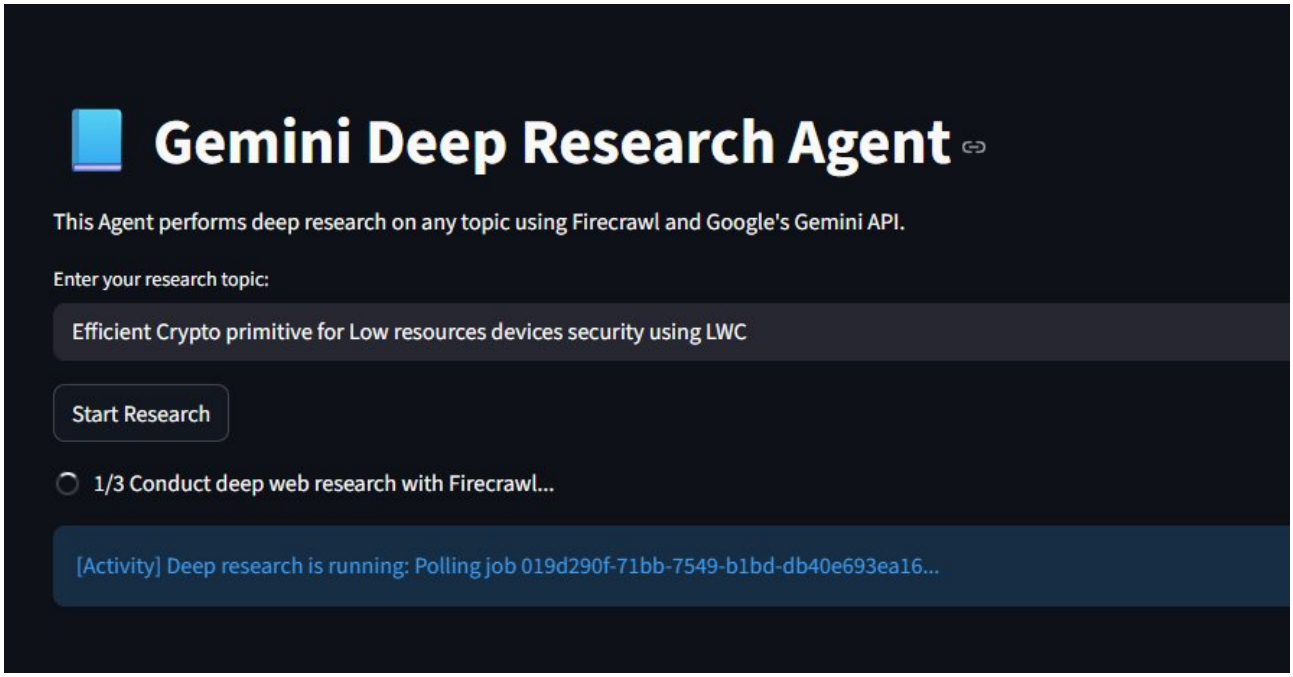

Figure 1: Step 1/3 – Conducting deep web research with Firecrawl

Figure 2: Step 2/3 – Synthesizing initial report with Gemini

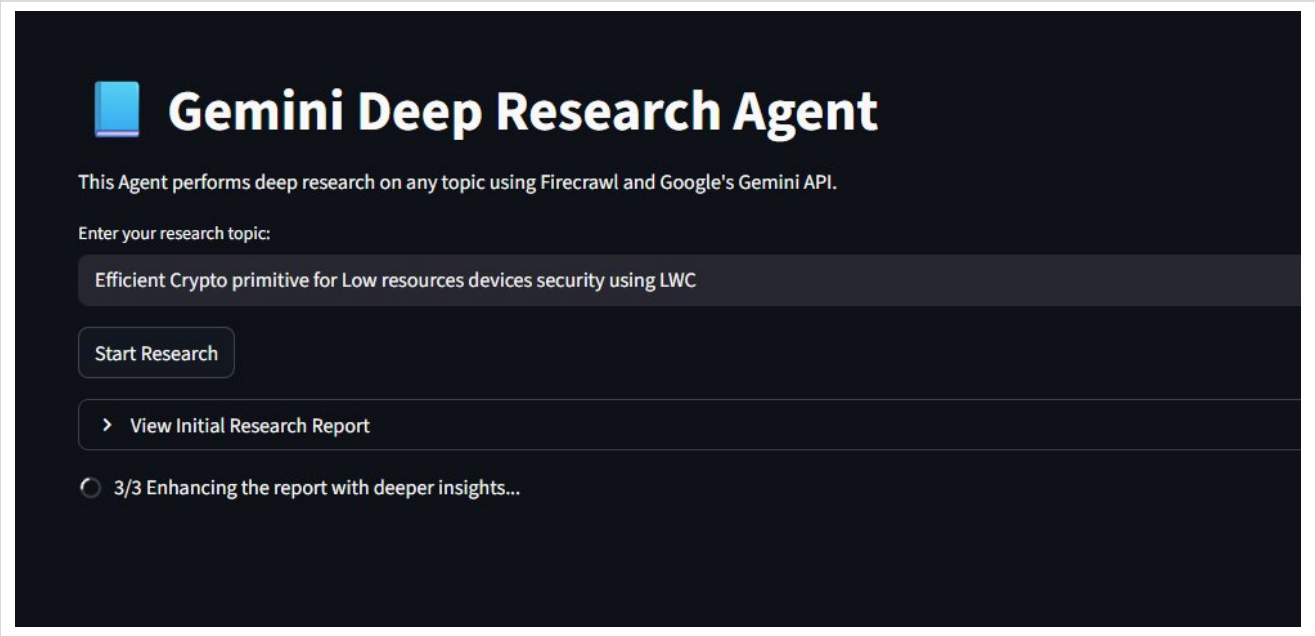

Figure 3: Step 3/3 – Enhancing the report with deeper insights

View Initial Research Report

Efficient Lightweight Cryptography for Low-Resource Device Security Using LWC

Executive Summary

This report delves into the critical role of Lightweight Cryptography (LWC) in securing the rapidly expanding landscape of low-resource Internet of Things (IoT) devices, particularly within industrial control systems (ICS) and healthcare applications. Traditional cryptographic algorithms are often too computationally intensive for these constrained environments, necessitating specialized LWC solutions. The report analyzes various LWC schemes, including innovative hybrid encryption algorithms based on chaotic maps and Fibonacci Q-matrices, alongside lightweight block and stream ciphers. It evaluates their performance using metrics such as NPCR, UACI, and entropy, assesses their security against cryptanalytic attacks, and provides a comparative analysis of different approaches. **Key findings indicate that hybrid chaotic systems offer robust security properties suitable for mission-critical applications, while highly optimized, simpler chaotic methods cater to extreme resource limitations, albeit with moderate security trade-offs.** The report emphasizes the need for LWC solutions that align with evolving industrial security standards and provide strong resistance against advanced threats.

1. Introduction

The ubiquitous growth of IoT devices, from smart healthcare sensors to industrial equipment and RFID tags, has revolutionized various sectors. However, this proliferation introduces significant cybersecurity vulnerabilities, especially given the inherent resource constraints of many IoT platforms (e.g., limited memory, processing power, and energy). Conventional cryptographic algorithms, such as AES or RSA, are often impractical for these devices due to their high computational overhead, which can impede real-time operations and drain limited battery life.

Lightweight Cryptography (LWC) has emerged as a crucial countermeasure, providing efficient techniques that uphold core security properties—confidentiality, integrity, and authenticity—with a significantly reduced computational footprint. This report comprehensively investigates a spectrum of lightweight cryptographic primitives. It evaluates their performance using standard metrics like Number of Pixel Change Rate (NPCR), Unified Averaged Change Intensity (UACI), entropy, and cross-entropy, while also reviewing their resilience against various cryptanalytic attacks, including differential analysis, side-channel, and brute-force attacks.

Digital security in industrial and healthcare IoT applications is further complicated by the imperative to comply with established standards (e.g., NIST's LWC standard, ISO/IEC 29192) and to maintain overall robustness against dynamic threat landscapes. This analysis highlights recent contributions, including new encryption paradigms built upon chaotic dynamics, methods integrating Fibonacci sequences with chaotic maps, and hybrid frameworks that optimally balance encryption strength with hardware efficiency.

Figure 4: Initial Research Report – Executive Summary & Introduction

2. Background and Literature Overview

2.1 Internet of Things and Resource Constraints

IoT devices, frequently built upon platforms like ARM Cortex-M microcontrollers, typically operate with minimal resources—often tens of kilobytes of memory and stringent energy budgets. Consequently, deploying full-scale conventional cryptographic solutions can severely degrade performance and battery life. LWC algorithms are specifically designed to mitigate these issues by reducing gate count, energy consumption, computational latency, and memory overhead, all while preserving an acceptable level of security.

2.2 Definitions

Key definitions shaping this discussion include:

- **Low-Resource Devices**: Platforms characterized by limited processing capability, memory (RAM/ROM), and energy, such as RFID tags and sensor nodes [1].
- **Lightweight Cryptography (LWC)**: Cryptographic techniques specifically optimized for resource-constrained environments. These typically feature key sizes between 80 to 128 bits and minimized rounds (6–40) to balance security with performance [2][3].
- **Cryptographic Primitives for LWC**: Encompass symmetric algorithms (block ciphers and stream ciphers), hash functions, and message authentication codes re-engineered for minimal computational overhead [4].
- **Synchronous vs. Asynchronous Cryptography**: Synchronous methods depend on time-synchronized operations between communicating parties, whereas asynchronous approaches operate independently, designed to tolerate timing uncertainties [5][6].

2.3 Literature Surveys and Research Gaps

Extensive literature reviews have compared numerous lightweight ciphers based on parameters such as block/key size, gate equivalent (GE) area, latency, and throughput [7][8][9]. While these surveys provide valuable comparisons, they often lack a holistic perspective integrating:

- **Industrial Security Standards**: The necessity of aligning with standards from NIST and ISO/IEC to ensure interoperability and compliance in critical infrastructures.
- **Robustness Against Advanced Attacks**: Detailed analysis of differential, linear, related-key, and side-channel attacks specifically tailored for the Industrial Internet of Things (IIoT).
- **Real-world Implementation Feasibility**: Practical evaluation of energy consumption and processing latency on common IoT platforms like Arduino and Raspberry Pi.

This report addresses these gaps by comparing emerging lightweight cryptographic techniques with established solutions, focusing particularly on their applicability in ICS and healthcare environments.

Figure 5: Research Report – Background & Literature Overview

References

The in-text citations, e.g., `[1]`, `[10]`, refer to the categorized bibliography provided in the original research context. For specific URLs of the sources used to compile this report, please refer to the list below.

[A] **Enhancing IoT security with resource-efficient cryptography.** *ScienceDirect.* Available at: https://www.sciencedirect.com/science/article/abs/pii/S1574013725001376

[B] **Security primitives for memoryless IoT devices based on Physical...** *Nature.* Available at: https://www.nature.com/articles/s41598-024-75373-6

[C] **NIST Finalizes 'Lightweight Cryptography' Standard to Protect Small...** *National Institute of Standards and Technology (NIST).* Available at: https://www.nist.gov/news-events/news/2025/08/nist-finalizes-lightweight-cryptography-standard-protect-small-devices

[D] **Efficiency and Security Evaluation of Lightweight Cryptographic...** *PMC (PubMed Central).* Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC11207458/

[E] **Efficient lightweight cryptographic solutions for enhancing data...** *Frontiers in Computer Science.* Available at: https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1522184/full

[F] **A lightweight cryptography (LWC) framework to secure memory...** *ScienceDirect.* Available at: https://www.sciencedirect.com/science/article/pii/S1110016820305858

[G] **A Comprehensive Review on Lightweight Cryptographic...** *ACM Digital Library.* Available at: https://dl.acm.org/doi/full/10.1145/3757734

[H] **Lightweight Cryptography for IoT Platforms: Enhancing Security in...** *IEEE Xplore.* Available at: https://ieeexplore.ieee.org/document/10913623/

[I] **LIGHT WEIGHT CRYPTOGRAPHY (LWC) ALGORITHMS IN TERMS...** *Mililink.* Available at: https://www.mililink.com/upload/article/458062550aams_vol_221_november_2022_a11_p127-137_v._muthu_ganeshan_and_s._sivagurunathan.pdf

3/3 Enhancing the report with deeper insights...

Figure 6: Research Report – References Section

5. Comparative Analysis and Recommendations

A quantitative comparison of several proposed lightweight cryptographic methods highlights their inherent trade-offs between security and resource efficiency:

Metric	Method 1: 4D Dynamic Chaos	Method 2: Hybrid Chaos (Logistic-Parity)	Method 3: Fibonacci-Chaotic Map
Entropy	11.1254	10.0521	8.1325
NPCR (%)	≈99.6151	≈99.6150	≈99.5775
UACI (%)	38.8925	37.3752	35.4550
Hardware Complexity	Moderate – high (due to 4D system)	Moderate – balanced	Low – simplest computation
Recommended Use	High-security applications (mission-critical)	Balanced scenarios where both security and resource efficiency matter	Resource-extreme environments with moderate security requirements; less ideal when high randomness is imperative

Interpretation:

- **Method 1 (4D Dynamic Chaos)** provides the highest security, demonstrated by its superior entropy and diffusion (NPCR & UACI) scores. **This method is best suited for mission-critical IIoT applications where robust security is paramount, despite requiring a moderately higher hardware budget.**
- **Method 2 (Hybrid Chaos - Logistic-Parity)** strikes an excellent balance between security and efficiency. **It is an ideal choice for scenarios where resource constraints are present but do not prohibit the moderate complexity required to achieve robust security.**
- **Method 3 (Fibonacci-Chaotic Map)** is distinguished by its extreme lightweight nature and minimal computational overhead. However, its lower entropy scores suggest it may not offer sufficient protection for environments demanding stringent security. **This method is most appropriate for resource-extreme environments where moderate security requirements are acceptable, but it is less ideal when high randomness and strong security are critical.**

Beyond these specific chaotic methods, standardized block and stream ciphers such as AES, ASCON, Camellia, CLEFIA, and HIGHT have also been evaluated. While these algorithms are well-established and proven secure, many demand substantial gate equivalents (GE) or incur higher latency, making them less suitable when device resources are severely limited.

Figure 7: Research Report – Comparative Analysis & Recommendations